

PRINCIPLES OF ENVIRONMENTAL PHYSICS

PRINCIPLES OF ENVIRONMENTAL PHYSICS

3rd Edition

JOHN MONTEITH

Edinburgh, United Kingdom

MIKE UNSWORTH

Corvallis, Oregon, United States



ELSEVIER

AMSTERDAM • BOSTON • HEIDELBERG • LONDON
NEW YORK • OXFORD • PARIS • SAN DIEGO
SAN FRANCISCO • SINGAPORE • SYDNEY • TOKYO

Academic Press is an imprint of Elsevier



Publishing Editor: Jeanne Lawson
Production Director: Lori A. Buck
Project Manager: Lori A. Buck
Manufacturing Manager: André A. Cuello

Academic Press is an imprint of Elsevier
30 Corporate Drive, Suite 400, Burlington, MA 01803, USA
525 B Street, Suite 1900, San Diego, California 92101-4495, USA
84 Theobald's Road, London WC1X 8RR, UK

This book is printed on acid-free paper. (∞)

Copyright © 2008, Elsevier Inc. All rights reserved.

No part of this publication may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopy, recording, or any information storage and retrieval system, without permission in writing from the publisher.

Permissions may be sought directly from Elsevier's Science & Technology Rights Department in Oxford, UK: phone: (+44) 1865 843830, fax: (+44) 1865 853333, E-mail: permissions@elsevier.com. You may also complete your request online via the Elsevier homepage (<http://elsevier.com>), by selecting "Support & Contact" then "Copyright and Permission" and then "Obtaining Permissions."

Library of Congress Cataloging-in-Publication Data
Application Submitted

British Library Cataloguing-in-Publication Data
A catalogue record for this book is available from the British Library.

ISBN: 978-0-12-505103-3

For information on all Academic Press publications
visit our Web site at www.books.elsevier.com

Printed in the United States of America
08 09 10 9 8 7 6 5 4 3 2 1

Working together to grow
libraries in developing countries

www.elsevier.com | www.bookaid.org | www.sabre.org

ELSEVIER

BOOK AID
International

Sabre Foundation

Contents

Preface	ix
Acknowledgments	xiii
Symbols	xvi
1 The Scope of Environmental Physics	1
2 Properties of Gases and Liquids	4
Gases and Water Vapor	4
Liquid	20
Stable Isotopes	24
Problems	25
3 Transport of Heat, Mass, and Momentum	26
General Transfer Equation	26
Molecular Transfer Processes	28
Diffusion Coefficients	32
Problems	37
4 Transport of Radiant Energy	38
The Origin and Nature of Radiation	38
Spatial Relations	44
Problems	51
5 Radiation Environment	52
Solar Radiation	52
Attenuation of Solar Radiation in the Atmosphere	57
Solar Radiation at the Ground	62
Terrestrial Radiation	73

Net Radiation	79
Problems	84
6 Microclimatology of Radiation (i)	86
Radiative Properties of Natural Materials	86
Problems	99
7 Microclimatology of Radiation (ii)	100
Geometric Principles	101
Diffuse Radiation	110
Problems	115
8 Microclimatology of Radiation (iii)	116
Interception of Radiation by Plant Canopies	116
Interception of Radiation by Animal Coats	133
Net Radiation	137
Problems	139
9 Momentum Transfer	142
Boundary Layers	142
Momentum Transfer to Natural Surfaces	147
Lodging and Windthrow	155
Problems	159
10 Heat Transfer	160
Convection	160
Measurements of Convection	166
Conduction	179
Insulation	181
Problems	188
11 Mass Transfer (Gases and Water Vapor)	190
Non-Dimensional Groups	191
Measurements of Mass Transfer	193
Ventilation	197
Mass Transfer through Pores	200
Coats and Clothing	208
Problems	210
12 Mass Transfer (Particles)	212
Steady Motion	212

Non-Steady Motion	214
Particle Deposition	215
Problems	227
13 Steady State Heat Balance (i)	229
Heat Balance Equation	230
Heat Balance of Thermometers	232
Heat Balance of Surfaces	238
Developments from the Penman Equation	250
Problems	256
14 Steady State Heat Balance (ii)	258
Heat Balance Components	259
The Thermo-Neutral Diagram	266
Specification of the Environment	270
Case Studies	272
Sheep	273
Problems	281
15 Transient Heat Balance	283
Time Constant	283
General Cases	285
Heat Flow in Soil	290
Problems	298
16 Micrometeorology (i)	300
Turbulent Transfer	301
Flux-Gradient Methods	309
Methods for Indirect Measurements of Flux above Canopies	323
Relative Merits of Methods of Flux Measurement	328
Turbulent Transfer in Canopies	329
Density Corrections to Flux Measurements	331
Problems	332
17 Micrometeorology (ii)	335
Resistance Analogues	335
Case Studies	343
Transport within Canopies	359
Problems	365
References	367

Bibliography	390
Appendix A	394
Solutions to Selected Problems	401
Index	414



Preface to the Third Edition

In the time since the first edition of *Principles of Environmental Physics* was published in 1973, the subject has developed substantially; indeed, many users of the first and second editions have contributed to the body of research that makes this third edition larger than the first. From the start, this text has been aimed at two audiences: undergraduate and graduate students seeking to learn how the principles of physics can be applied to study the interactions between plants and animals and their environments; and the research community, particularly those involved in multidisciplinary environmental research. In many ways, environmental physics has become more thoroughly embedded in environmental research over the decades. For example, in ecology and hydrology, concepts of atmospheric exchange of gases and energy between organisms and the atmosphere, and the resistances (or conductances) controlling them are commonly applied. And in atmospheric science, soil-vegetation-atmosphere transfer schemes (SVATS) are an integral part of general circulation, mesoscale, and climate models. This “union of ideas” across the disciplines has made it challenging to define the scope of this third edition and to keep the size manageable. In doing so we have been guided by the word *Principles* in the title, so have focused, as in previous editions, on describing the critical principles of energy, mass, and momentum transfer, and illustrating them with a number of examples of their applications, taken from a range of classic and more recent publications.

Several themes have waxed and waned over the editions. At the time of the first edition, agricultural crop micrometeorology was a dominant

application of environmental physics, beginning with the desire to quantify the water use and irrigation requirements of crops, and extending, as new instrumentation became available, to the analysis of carbon dioxide exchange in efforts to identify the environmental controls of crop productivity. There has been much less new work on agricultural crop micrometeorology in the last decade or two, but applications of environmental physics to the study of managed and natural forests and other ecosystems gathered pace through the 1970s and 1980s, and these topics, rather than agricultural applications, probably currently account for the major fraction of the annual reviewed publications in environmental physics. Also in the 1970s and 1980s, concern grew over human influences on air quality (particularly acid rain and ozone), leading to the application of environmental physics to study fluxes of pollutant gases, acidic particles, and mist to crops and forests. Additionally, the technology for remote sensing from satellites developed considerably.

The second edition of *Principles of Environmental Physics*, published in 1990, reflected these developments by adding a new chapter on particle transfer, new material on radiative transfer, and expanding the sections on micrometeorological methods. It also expanded treatment of the environmental physics of animals and their environments, influenced by the work of a number of researchers studying the heat balance of livestock and wild animals, who began to use the terminology of environmental physics, thus establishing parallels with the integration of environmental physics into plant science. The identification of the ozone hole above Antarctica in 1985, and its influence on ultraviolet radiation at the surface, received a short mention in the second edition, but emerging research on deposition of nitrogen-containing gases to vegetation was not covered; both topics receive more attention in this edition.

Through the 1980s, extending to the present time, concern over rising concentrations of carbon dioxide and other greenhouse gases in the atmosphere and consequent likely effects on climate has been a dominant topic, leading to an explosion of measurement and modeling research programs that make use of principles described in this book. Two developments have been particularly important: improved instrumentation allowing the eddy covariance technique in micrometeorology to be applied for studies of land-atmosphere exchange of carbon dioxide, water vapor and some other trace gases over seasonal and multi-annual periods; and theoretical advances to enable models of plant-atmosphere exchange to be scaled up from the leaf scale to landscape, regional, and even global scales, creating links between the principles described in this book at organism and

canopy levels with the type of regional and global modeling necessary to address climate-change concerns. This edition contains two substantially revised chapters on micrometeorology with expanded treatment of the eddy covariance method, and with several new case studies to illustrate the application of micrometeorological methods over forests and natural landscapes. We have also expanded the material on solar and terrestrial radiation with new discussion of the roles of radiatively active greenhouse gases and aerosols.

Although eddy covariance has become the method of choice for micrometeorology in many situations, we have retained the material describing profile (similarity) techniques for deducing fluxes, because an understanding of similarity methods is essential for large-scale models and because profile methods have advantages in terms of simplicity of instrumentation when designing student projects or working with limited resources.

Other changes in this edition resulted from our own experience and feedback from those using this book as a teaching text: several sections that were particularly condensed in earlier editions have been expanded to aid clarity, and some sections have been rearranged to improve the flow; more worked examples have been included in the text; some specialized material (for example, details of the physics of radiative emission, and of radiation interaction with aerosols) has been added in text boxes that can be omitted by readers seeking a briefer treatment of the subject; and numerical problems have been added at the end of each chapter. Many of the numerical problems are more extensive than typically found in text books. This reflects requests we have had over the years from teachers who would like to explore realistic applications of the subject; many of the problems have been used in our own undergraduate and graduate teaching at Nottingham and Oregon State Universities, and we thank many students for their feedback and suggestions for improvements to the problems.

In planning this third edition, we debated whether to change nomenclature in flux equations from resistances to conductances, and whether to express quantities in “mole” units rather than “m·kg·s” units. Biologists increasingly use conductances and moles in their analyses, and there are some good theoretical and didactic arguments for this. However, we preferred to retain the “resistance” terms and “s m⁻¹” units used in earlier editions: the analogy with Ohm’s Law emphasizes the underlying physics of many analyses used in this book, and units of s m⁻¹ for resistances are more intuitive for heat and mass transfer calculations in energy balance and hydrological applications. Nevertheless, recognizing that many readers will be familiar with conductance and mol units, we have discussed

conversions of units at several appropriate points in the text. There are many advantages in environmental physicists becoming familiar with both systems of units to facilitate communication across the disciplines.

We intend this text to be useful for teaching undergraduates and graduate students specializing in physics, biology, and the environmental sciences. The mathematical treatment is deliberately kept relatively simple, with little use of calculus; the biology is also strictly limited, consisting principally of material essential for understanding the physical applications. There is a bibliography directing readers to more detailed texts if necessary. For our other category of readers, research scientists, we have continued the approach of previous editions by including a large number of references to peer-reviewed literature, identifying a mix of papers that we consider classics and ground-breaking research applications; about 33% of the references in this edition have been published since 1990.

In the Preface to the second edition we expressed the hope that our book would encourage more university physics departments to expose their students to environmental physics. Our impression is that progress has been slow. This surely cannot be because of a lack of career opportunities—current environmental concerns open many possibilities for environmental physicists in the atmospheric sciences, hydrology, ecology, and biology, particularly if they enjoy the challenges of multidisciplinary work. Nor does it seem to be because physics students lack interest in environmental subjects. Perhaps it is inevitable that the crowded physics curriculum leaves little room for options such as environmental physics, but it would be satisfying if, by the time the fourth edition of this book appears, environmental physics became as common as astronomy or meteorology as an optional course in physics departments.

John Monteith and Mike Unsworth, 2007



Acknowledgments

We thank the following for allowing us to use diagrams, photographs, and original data:

Dr C.D. Keeling and the database provided by the U.S. Department of Energy through its Carbon Dioxide Information Analysis Center at Oak Ridge National Laboratory (Figure 2.2);

Dr. J. Lean for providing the data used in Figure 5.1; Dr. S.T. Henderson and Adam Hilger Publishers (5.2);

Dr. J.A Coakley (5.3); NASA Goddard SpaceFlight Center for the TOMS data in Figure 5.4; The Solar Energy Research Institute for the computer models used to construct Figs. 5.5;

Dr. M.D. Steven and the Royal Meteorological Society (5.6) from the *Quarterly Journal of the Royal Meteorological Society*;

Dr. F. Vignola for providing the data used in Figure 5.8;

Dr. J.V. Lake for providing the data used in Fig 5.9;

Mr. F.E. Lumb and the Royal Meteorological Society (5.10) from the *Quarterly Journal of the Royal Meteorological Society*;

Dr. R. von Fleischer and the Deutschen Wetterdienstes (5.15);

Dr R. Nakamura for providing the data used in Fig 5.17; Dr. K. Bible for providing the data used in Fig 5.18;

Dr. E.L. Deacon and Elsevier Publishing Co. (6.1 and 15.7);

Dr. S.A. Bowers and Lippincott Williams and Wilkins Co. (6.3) from *Soil Science*;

Dr. K.J. McCree and Elsevier Publishing Co. (6.4) from *Agricultural Meteorology*;

Dr. G. Stanhill and Pergamon Press (6.6) from *Solar Energy*;
 Professor L.E. Mount and Edward Arnold (6.7, 14.3 and 14.4);
 Dr. J.C.D. Hutchinson and Pergamon Press (6.8) from *Comparative Biochemistry and Physiology*;
 Dr. W. Porter for providing data used in Figure 6.9;
 Dr. C.R. Underwood and Taylor and Francis Ltd (7.5) from *Ergonomics*;
 Dr. G.S. Campbell and Nottingham University Press (8.3);
 Dr. K. Cena and the Royal Society of London (8.6) from the *Proceedings of the Royal Society*;
 Dr. J. Grace and Oxford University Press (9.3) from *Journal of Experimental Botany*; The Royal Meteorological Society (9.4, 9.5, 13.6, 13.7, 13.8, 13.9, 15.3, 17.13 and 17.14) from the *Quarterly Journal of the Royal Meteorological Society*;
 Dr. W.C. Hinds and John Wiley & Sons Inc. (9.6, 12.7, 12.8, 12.9, 12.10);
 Dr. D. Aylor and the American Society of Plant Physiologists (9.7) from *Plant Physiology*;
 Dr. A Stokes and Cambridge University Press (9.8);
 Drs. C.J. Wood and R. Belcher and D. Reidel Publishing Co. (9.10) from *Boundary Layer Meteorology*;
 Dr. J.A. Clark and D. Reidel Publishing Co. (10.3) from *Boundary Layer Meteorology*;
 Dr. B.J. Bailey and the International Society for Horticultural Science (10.4) from *Acta Horticulturae*;
 Dr. S. Vogel and Clarendon Press (10.5);
 Dr. A.J. McArthur and the Royal Society of London (10.7, 10.8) from the *Proceedings of the Royal Society*;
 Dr. R.P. Clark and *The Lancet* (10.9, 10.11) and Cambridge University Press (10.8) from *Journal of Physiology*;
 Dr. P.F. Scholander and the Marine Biological Laboratory (10.12);
 Dr. I. Impens for allowing us to use unpublished measurements in Figure 11.1.
 Dr. T. Haseba and the Society of Agricultural Meteorology in Japan (11.3) from *Journal of Agricultural Meteorology*;
 Dr. H.G. Jones and Cambridge University Press (11.7);
 Dr. D. Aylor and Pergamon Press (12.5) from *Atmospheric Environment*;
 Professor N.A. Fuchs and Pergamon Press (12.1);
 Dr. A.C. Chamberlain and Academic Press (12.3), and D. Reidel Publishing Co. (17.1) from *Boundary Layer Meteorology*;

Dr. P. Little and Pergamon Press (12.6) from *Atmospheric Environment*;

Dr. K. Raschke and Springer-Verlag (13.5) from *Planta*;

R. Milstein (13.10); Elsevier Publishing Co (13.11, 13.12) from the *Journal of Hydrology* and (17.11) from *Agricultural and Forest Meteorology*;

Dr. A.M. Hemmingsen (14.2);

Dr. D.M. Gates and Springer-Verlag (15.2);

Dr. J. van Eimern and the Deutschen Wetterdienstes (15.5);

Dr. W.R. van Wijk and North Holland Publishing Co. (15.6);

D. Vickers and Dr. L. Mahrt (16.3);

Dr. J. Finnigan and D. Reidel Publishing Co. (16.4) from *Boundary Layer Meteorology* and (17.16, 17.17);

Dr. R.H. Shaw and Elsevier Publishing Co. (16.8) from *Agricultural Meteorology*; Academic Press (16.9 and 16.10);

Dr. M.R. Raupach and Annual Reviews Inc. (16.11) from *Annual Review of Fluid Mechanics*;

Dr. T.A. Black and Blackwell Scientific (17.9, 17.10) from *Global Change Biology*;

Dr. D. Baldocchi (17.7) and D. Reidel Publishing Co. (17.19) from *Boundary Layer Meteorology*;

Dr. M. Sutton and the Royal Society (17.15) from the *Philosophical Transactions of the Royal Society of London*.

We particularly thank the staff at ChapterTwo for their skill and patience in converting our text, prepared in Scientific Word, into this book.



Symbols

The main symbols used in this book are arranged here in a table containing brief definitions of each quantity. A few of the symbols are universally accepted (e.g. R , g), some have been chosen because they appear very frequently in the literature of environmental physics (e.g., r_s , z_o , K_M), and some have been devised for the sake of consistency. In particular, the symbols **S** and **L** are used for flux densities of short-and long-wave radiation with subscripts to identify the geometrical character of the flux, e.g., **S_d** for the flux of diffuse short-wave radiation from the sky.

Flux densities of momentum, heat and mass are printed in bold case throughout the book (e.g., τ , **E**) and so is the latent heat of vaporization of water λ , partly to distinguish it from wavelength λ and partly because it is often associated with **E**. Upper case subscripts are used to refer to momentum, heat, vapor, carbon dioxide, etc., e.g., r_V , K_M ; most other subscripts are lower case, e.g., c_p for the specific heat of air at constant pressure.

The complete set of symbols represents the best compromise that could be found between consistency, clarity and familiarity.

ROMAN ALPHABET

A	area; azimuth angle with respect to south
A_b	area of solid object projected on a horizontal plane
A_p	area of solid object projected on plane perpendicular to solar beam
$A(z)$	amplitude of soil temperature wave at depth z
B	total energy emitted by unit area of full radiator or black body

B	wet-bulb depression
$\mathbf{B}(\lambda)$	energy per unit wavelength in spectrum of full radiator or black body
c	volume fraction at CO_2 (e.g., v.p.m.); fraction of sky covered by cloud; velocity of light; mean velocity of gas molecules
c_d	drag coefficient for form drag and skin friction combined
c_f	drag coefficient for form drag
c_l	specific heat of a liquid
c_p	specific heat of air at constant pressure; efficiency of impaction of particles
c_s	specific heat of solid fraction of a soil
c_v	specific heat at constant volume
c'	bulk specific heat of soil
C	flux of heat per unit area by convection in air
$\mathcal{C}$	heat capacity of an organism per unit surface area
d	zero plane displacement
D	saturation vapor pressure deficit; diffusion coefficient for a gas in air (subscripts V for water vapor; C for CO_2); damping depth $(2\kappa'/\omega)^{1/2}$
e	partial pressure of water vapor in air
$e_s(T)$	saturation vapor pressure of water vapor at temperature T
δe	saturation deficit, i.e., $e_s(T) - e$
E_q	energy of a single quantum
$\mathbf{E}$	flux of water vapor per unit area; evaporation rate
$\mathbf{E}_r$	respiratory evaporation rate of animal
$\mathbf{E}_s$	rate of evaporation from skin
$\mathbf{E}_t$	rate of evaporation from vegetation
F	generalized stability factor $(\phi_v\phi_m)^{-1}$
F	drag force on a particle; retention factor
$\mathbf{F}$	mass flux of a gas per unit area; flux of radiant energy
g	acceleration by gravity (9.81 m s^{-2})
$\mathbf{G}$	flux of heat by conduction, per unit area
h	Planck's constant ($6.63 \times 10^{-34} \text{ J s}$); relative humidity of air; height of cylinder, crop, etc.
$\mathbf{H}$	total flux of sensible and latent heat, per unit area
i	intensity of turbulence, i.e., root mean square velocity/mean velocity
$\mathbf{I}$	intensity of radiation (flux per unit solid angle)

J	rate of change of stored heat per unit area
<i>k</i>	von Karman's constant (0.41); thermal conductivity of air; attenuation coefficient; Boltzmann constant ($1.38 \times 10^{-23} \text{ J K}^{-1}$)
<i>k'</i>	attenuation coefficient; thermal conductivity of a solid
<i>K</i>	diffusion coefficient for turbulent transfer in air (subscripts H for heat, M for momentum, V for water vapor, C for CO ₂)
$\mathcal{H}$	canopy attenuation coefficient
$\mathcal{H}_s$	ratio of horizontally projected area of an object to its plane or total surface area
<i>l</i>	mixing length, stopping distance; length of plate in direction of airstream
<i>L</i>	leaf area index; Monin–Obukhov length
<i>L</i>	flux of long-wave radiation per unit area (subscript u for upwards; d for downwards; e from environment; b from body)
<i>m</i>	mass of a molecule or particle; air mass number
<i>M</i>	rate of heat production by metabolism per unit area of body surface
<i>M</i>	grammolecular mass (subscripts a for dry air, v for water vapor)
<i>n</i>	represents a number or dimensionless empirical constant in several equations
<i>N</i>	Avogadro constant (6.02×10^{23}); number of hours of daylight
<i>N</i>	radiance (radiant flux per unit area per unit solid angle)
P	latent heat equivalent of sweat rate per unit body area
<i>p</i>	total air pressure; interception probability in hair coats
<i>q</i>	specific humidity of air (mass of water vapor per unit mass of moist air)
<i>Q</i>	rate of mass transfer
<i>r</i>	radius; resistance to transfer (subscripts M momentum, H heat, V water vapor, C for CO ₂); usually applied to boundary layer transfer
<i>r_a</i>	resistance to transfer in the atmosphere (subscripts M, H, V, C as above)
<i>r_b</i>	boundary layer resistance of crop for mass transfer
<i>r_c</i>	canopy resistance
<i>r_d</i>	thermal resistance of human body
<i>r_f</i>	thermal resistance of hair, clothing; resistance for forced ventilation in open-top chambers
<i>r_p</i>	resistance of pore for mass transfer
<i>r_s</i>	resistance of a set of stomata

r_h	resistance of hole (one side) for mass transfer
r_i	incursion resistance for open-top chambers
r_t	total resistance of single stoma
r_H	resistance for heat transfer by convection, i.e., sensible heat
r_R	resistance for radiative heat transfer ($\rho c_p / 4\sigma T^3$)
r_{HR}	resistance for simultaneous sensible and radiative heat exchange, i.e., r_H and r_R in parallel
r_V	resistance for water vapor transfer
R	Gas Constant ($8.31 \text{ J mol}^{-1} \text{ K}^{-1}$)
R_n	net radiation flux density
R_{ni}	net radiation absorbed by a surface at the temperature of the ambient air
s	amount of entity per unit mass of air
S	gas concentration
S_d	diffuse solar irradiance on horizontal surface
S_e	solar radiation received by a body, per unit area, as a result of reflection from the environment
S_p	direct solar irradiance on surface perpendicular to solar beam
S_b	direct solar irradiance on horizontal surface
S_t	total solar irradiance (usually) on horizontal surface
t	diffusion pathlength
T	temperature
T_a	air temperature
T_b	body temperature
T_c	cloud-base temperature
T_d	dew point temperature
T_e	equivalent temperature of air ($T + (e/\gamma)$)
T_{e^*}	apparent equivalent temperature of air ($T + (e/\gamma^*)$)
T_f	effective temperature of ambient air
T_s, T_0	temperature of surface losing heat to environment
T_v	virtual temperature
T'	thermodynamic wet bulb temperature
T^*	standard temperature for vapor pressure specification
u	optical pathlength of water vapor in the atmosphere
$u(z)$	velocity of air at height z above earth's surface
u_*	friction velocity

v	molecular velocity
v_d	deposition velocity
v_s	sedimentation velocity
V_m	molar volume at STP (22.4 L)
$\dot{V}$	volume
w	vertical velocity of air; depth of precipitable water
W	body weight of animal
x	volume fraction (subscripts s for soil; l for liquid; g for gas); ratio of cylinder height to radius
z	distance; height above earth's surface
z_o	roughness length
Z	height of equilibrium boundary layer

GREEK ALPHABET

α	absorption coefficient (subscripts p for photosynthetically active; T for total radiation; r for red; i for infrared)
$\alpha(\lambda)$	absorptivity at wavelength λ
β	solar elevation; ratio of observed Nusselt number to that for a smooth plate
γ	psychrometer constant ($= c_p p / \lambda \varepsilon$)
γ^*	apparent value of psychrometer constant ($= \gamma r_v / r_H$)
Γ	dry adiabatic lapse rate, DALR ($9.8 \times 10 \text{ k m}^{-1}$)
δ	depth of a boundary layer
Δ	rate of change of saturation vapor pressure with temperature, i.e., $\partial e_s(T) / \partial T$
ε	ratio of molecular weights of water vapor and air (0.622)
ε_a	apparent emissivity of the atmosphere
$\varepsilon(\lambda)$	emissivity at wavelength λ
θ	angle with respect to solar beam; potential temperature
κ	thermal diffusivity of still air
κ'	thermal diffusivity of a solid, e.g., soil
λ	wavelength of electromagnetic radiation
λ	latent heat of vaporization of water
μ	coefficient of dynamic viscosity of air
ν	coefficient of kinematic viscosity of air; frequency of electromagnetic radiation

ρ	reflection coefficient (subscripts p for photosynthetically active; c for canopy; s for soil; r for red; i for infrared; T for total radiation); density of a gas, e.g., air including water vapor component
ρ_a	density of dry air
ρ_c	density of CO ₂
ρ_l	density of a liquid
ρ_s	density of a solid component of soil
ρ'	bulk density of soil
ρ	reflection coefficient
$\rho(\lambda)$	reflectivity of a surface at wavelength λ
σ	Stefan–Boltzmann constant ($5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$)
Σ	the sum of a series
τ	flux of momentum per unit area; shearing stress
τ	fraction of incident radiation transmitted, e.g., by a leaf; relaxation time, time constant, turbidity coefficient
ϕ	mass concentration of CO ₂ , e.g., g m^{-3} ; angle between a plate and airstream
Φ	radiant flux density
χ	absolute humidity of air
$\chi_s(T)$	saturated absolute humidity at temperature $T(^{\circ}\text{C})$
ψ	angle of incidence
ω	angular frequency; solid angle

NON-DIMENSIONAL GROUPS

Le	Lewis number (κ/D)
Gr	Grashof number
Nu	Nusselt number
Pr	Prandtl number (ν/κ)
Re _*	Roughness Reynolds number ($u_* z_o/\nu$)
Re	Reynolds number
Ri	Richardson number
Sc	Schmidt number (ν/D)
Sh	Sherwood number
Stk	Stokes number

LOGARITHMS

ln	logarithm to the base e
log	logarithm to the base 10